

## **Week 8 updates**

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## Attempted fixes

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- Log-normal time sampling to over-emphasize intermediate times - extremely unstable sampling when trained on small number of epochs, might work with a lot more
- Translation of  $x_1$  by +10 resulted in the model outputting 0
- Train from  $x_0 \sim \mathcal{N}(0, 1)$  but clamp to  $[-1, 1]$  - mostly reduces exploding outliers but results in low-diversity output

# Variable-sized clouds

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- Can't find bug in zero padding
- Potential fix: apply curriculum learning approach. Sort training data such that average jet size per batch decreases over time

# Potential fix for exploding output

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- Return  $x^L - x_0$  instead of  $x^L$  - when the model performs poorly and returns an identity output, we return 0 as opposed to a linearly exploding output

# Rotation invariance

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Apply approach of Song et al. 2023 to permute and rotate particles in  $x_0 \sim \mathcal{N}$  to align with  $x_1 \sim p_{\text{data}}$  in every  $(x_0, x_1)$  training pair

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**Algorithm 3** A variant of iterative closest point (ICP) algorithm.

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- 1: **Input:** a point cloud  $\mathbf{z} \in \mathbb{R}^{3 \times N}$  and its reference point cloud  $\mathbf{y} \in \mathbb{R}^{3 \times N}$ .
  - 2: **while**  $\tau$  has not converged **do**
  - 3:   Obtain permutation matrix  $\mathbf{\Pi} = \underset{\mathbf{\Pi}}{\operatorname{argmin}} \|\mathbf{\Pi}(\mathbf{R}\mathbf{z})^\top - \mathbf{y}^\top\|_2$  with Jonker-Volgenant algorithm [50]
  - 4:   Obtain rotation matrix  $\mathbf{R} = \underset{\mathbf{R}}{\operatorname{argmin}} \|\mathbf{R}(\mathbf{\Pi}\mathbf{z})^\top - \mathbf{y}^\top\|_2$  with Kabsch algorithm [51]
  - 5:    $\tau = \|\mathbf{\Pi}(\mathbf{R}\mathbf{z}^\top)^\top - \mathbf{y}^\top\|_2$
  - 6: **end while**
  - 7: **return**  $\mathbf{\Pi}, \mathbf{R}$
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# Boost invariance

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- Song et al. 2023 achieve translation invariance by projecting  $x_0$  to zero COM space during training, and projecting  $x_t$  to zero COM at every step of the ODE solver
- We have the target jet's 4-momenta, so we can boost the noisy particles  $x_0$  from their own reference frame to the reference frame of the jet at initialization

# Coordinate space

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Approach of Spinner et al. 2024:

- Polar coordinates are more physically meaningful than Cartesian coordinates, but working in Cartesian coordinates is better for equivariance
- The network works entirely in Cartesian coordinates, and its outputs are projected to polar coordinates
- Loss is calculated in terms of polar coordinates
- Should mitigate the zero-collapse issue

$$\begin{bmatrix} 0 & -\frac{px\,pz}{(px^2+py^2)^{\frac{3}{2}}\sqrt{\frac{pz^2}{px^2+py^2}+1}} & -\frac{py\,pz}{(px^2+py^2)^{\frac{3}{2}}\sqrt{\frac{pz^2}{px^2+py^2}+1}} & \frac{1}{\sqrt{px^2+py^2}\sqrt{\frac{pz^2}{px^2+py^2}+1}} \\ 0 & -\frac{py}{px^2+py^2} & \frac{px}{px^2+py^2} & 0 \\ 0 & \frac{px}{\sqrt{px^2+py^2}} & \frac{py}{\sqrt{px^2+py^2}} & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$